

VPA ICT — Grades 4–5 Term 2: Spreadsheets & Presentations

Term 2: Spreadsheets & Presentations — Weeks 13 – 24

Theme: Excel/Sheets basics, formulas, charts, PowerPoint/Slides | Duration: 40 minutes per lesson

Term Overview

Students learn to organize, calculate, and visualize data using spreadsheets. They then apply presentation skills to communicate findings effectively. By the end of this term, students can create a spreadsheet with formulas and charts, and deliver a short presentation with visual aids.

Week-by-Week Scheme of Work

Week 13 — Spreadsheet Introduction

Keywords: spreadsheet 电子表格 (diànzǐ biǎogé), cell 单元格 (dānyuán gé), row 行 (háng), column 列 (liè), worksheet 工作表 (gōngzuò biǎo), workbook 工作簿 (gōngzuò bù)

Activity 1 (15 min): "Excel/Sheets Exploration" — Students open a blank spreadsheet and identify rows, columns, cells, and the formula bar.

Activity 2 (15 min): "Cell Address Bingo" — Teacher calls out cell addresses (B3, D7); students click the correct cell.

Activity 3 (10 min): Data entry practice — Students type their name, age, and favorite number into cells A1, A2, A3.

Formative: Can identify cell addresses and enter data accurately.

Week 14 — Data Entry & Formatting

Keywords: format 格式 (géshì), currency 货币 (huòbì), percentage 百分比 (bǎifēnbǐ), date 日期 (rìqī), decimal 小数 (xiǎoshù)

Activity 1 (15 min): "Class Birthday Chart" — Students enter classmate birthdays into a spreadsheet with columns: Name, Month, Day. Format dates consistently.

Activity 2 (15 min): Formatting challenge — Apply bold headers, center alignment, alternating row colors.

Activity 3 (10 min): Sorting practice — Sort birthdays by month. Who has a birthday in January?

Formative: Spreadsheet formatted with headers, aligned data, sorted correctly.

Week 15 — Basic Formulas (+, -, *, /)

Keywords: formula 公式 (gōngshì), sum 求和 (qiúhé), calculate 计算 (jìsuàn), equals 等于 (děngyú), operator 运算符 (yùnsuàn fú)

Activity 1 (15 min): "Shopping List Calculator" — Students create a shopping list with item names, quantities, and prices. Calculate total cost using formulas (=B2*C2).

Activity 2 (15 min): Formula treasure hunt — Hidden formulas in a spreadsheet. Students find and fix 5 broken formulas.

Activity 3 (10 min): "What happens if...?" — Change a quantity. Watch the total update automatically. Magic!

Formative: Creates correct multiplication and addition formulas.

Week 16 — AVERAGE, MAX, MIN

Keywords: average 平均值 (píngjūn zhí), maximum 最大值 (zuìdà zhí), minimum 最小值 (zuìxiǎo zhí), range 范围 (fànweí)

Activity 1 (15 min): "Sports Score Analysis" — Students enter scores from a fictional sports tournament. Calculate average, highest, and lowest scores.

Activity 2 (15 min): "Class Height Chart" — Students anonymously enter heights. Find class average, tallest, shortest.

Activity 3 (10 min): Discussion — "What does AVERAGE tell us? What doesn't it tell us?"

Formative: Uses AVERAGE, MAX, MIN functions correctly.

Week 17 — Creating Bar Charts

Keywords: chart 图表 (túbiǎo), bar chart 柱状图 (zhùzhuàng tú), axis 轴 (zhóu), label 标签 (biāoqiān), legend 图例 (tú lì), title 标题 (biāotí)

Activity 1 (15 min): "Favorite Animals Survey" — Class votes on favorite animals. Students enter data and create a bar chart.

Activity 2 (15 min): Chart formatting — Add chart title, axis labels, data labels, and change colors.

Activity 3 (10 min): "Chart Detective" — Given a chart, what 3 things can you learn? Write sentences.

Formative: Bar chart has title, labeled axes, and clear data representation.

Week 18 — Pie & Line Charts

Keywords: pie chart 饼图 (bǐngtú), line chart 折线图 (zhéxiàn tú), percentage 百分比 (bǎifēnbǐ), trend 趋势 (qūshì), data point 数据点 (shùjù diǎn)

Activity 1 (15 min): "Weather Data Visualization" — Students enter weekly temperature data and create a line chart showing the trend.

Activity 2 (15 min): Pie chart of class preferences — Create a pie chart showing how students travel to school.

Activity 3 (10 min): "Which chart is best?" — Given 3 data sets, choose the best chart type for each.

Formative: Creates appropriate chart type with correct labels.

Week 19 — Introduction to Presentations

Keywords: presentation 演示文稿 (yǎnshì wénǎo), slide 幻灯片 (huàndēngpiàn), title slide 标题页 (biāotí yè), content slide 内容页 (nèiróng yè), slide layout 幻灯片版式 (huàndēngpiàn bǎnshì)

Activity 1 (15 min): "PowerPoint/Slides Basics" — Open software, add a new slide, choose layouts, type text.

Activity 2 (15 min): "My First Presentation" — 3 slides: Title (name), About Me (3 bullet points), My Favorite Thing (1 picture).

Activity 3 (10 min): Slide design principles — "Less text, more visuals!" "Big font, easy to read!"

Formative: Presentation has 3 slides with appropriate layouts.

Week 20 — Slide Design & Layouts

Keywords: theme 主题 (zhǔtí), template 模板 (múbǎn), consistent 一致的 (yīzhì de), color scheme 配色方案 (pèisè fāng'àn), contrast 对比度 (duìbǐdù)

Activity 1 (15 min): "Theme Challenge" — Apply 3 different themes to the same presentation. Vote on the best one.

Activity 2 (15 min): "Design Police" — Show good and bad slide designs. Students identify problems: too much text, bad colors, tiny font.

Activity 3 (10 min): Create a consistent 5-slide presentation with one theme.

Formative: Design is consistent and readable across all slides.

Week 21 — Animations & Transitions

Keywords: animation 动画 (dòng huà), transition 切换 (qiē huàn), entrance 进入 (jìn rù), emphasis 强调 (qiáng diào), exit 退出 (tuì chū)

Activity 1 (15 min): "Do's and Don'ts" — Show extreme examples: no animations vs. too many animations. Create class rules.

Activity 2 (15 min): Practice — Add ONE entrance animation per slide. Add ONE subtle transition between slides.

Activity 3 (10 min): "Animation with Purpose" — When should we use animation? (Reveal answers, focus attention, build suspense.)

Formative: Uses animations purposefully and sparingly.

Week 22 — Embedding Media

Keywords: embed 嵌入 (qiān rù), video 视频 (shì pín), hyperlink 超链接 (chāo liàn jiē), audio 音频 (yīn pín)

Activity 1 (15 min): "Video Presentation" — Students insert a short educational video into a slide and add discussion questions.

Activity 2 (15 min): Hyperlink practice — Create a slide with 3 hyperlinks to safe websites.

Activity 3 (10 min): Troubleshooting — "What if the video doesn't play?"

Week 23 — Presenting Skills

Keywords: present 演示 (yǎn shì), audience 观众 (guān zhòng), eye contact 眼神交流 (yǎn shén jiāo liú), speak clearly 清晰地说 (qīng xī de shuō), pace 语速 (yǔ sù)

Activity 1 (15 min): "Public Speaking Tips" — Video of good/bad presentations. Class discussion.

Activity 2 (20 min): Peer presentations — Groups of 3. Each student presents 2 slides. Peers give feedback using: "I liked..." "I suggest..."

Activity 3 (5 min): Self-reflection — "One thing I did well, one thing to improve."

Week 24 — Term 2 Assessment

Project Assessment (40 min):

Students complete a data-to-presentation project:

Create a spreadsheet with survey data (minimum 10 responses, 3 categories)

Use at least 2 formulas (SUM and AVERAGE)

Create 1 chart from the data

Create a 5-slide presentation about findings

Include chart image in presentation

Present 2-minute summary to a partner

Marking Criteria:

Bronze: Spreadsheet with data + 1 formula; 3 basic slides

Silver: Spreadsheet with 2+ formulas and 1 chart; 5 slides with design; 2-min presentation

Gold: All above + creative design, clear conclusion in presentation, confident delivery

Standards Mapping

Weeks	Standard	Description
13 – 18	CCSS. MATH. 4. OA. C. 5	Generate and analyze patterns
13 – 18	CSTA 1B-DA-07	Collect and analyze data using computational tools
19 – 23	ISTE 1.5	Communicate effectively using digital tools
19 – 23	CCSS. ELA. SL. 4. 4	Report on a topic using relevant, descriptive details